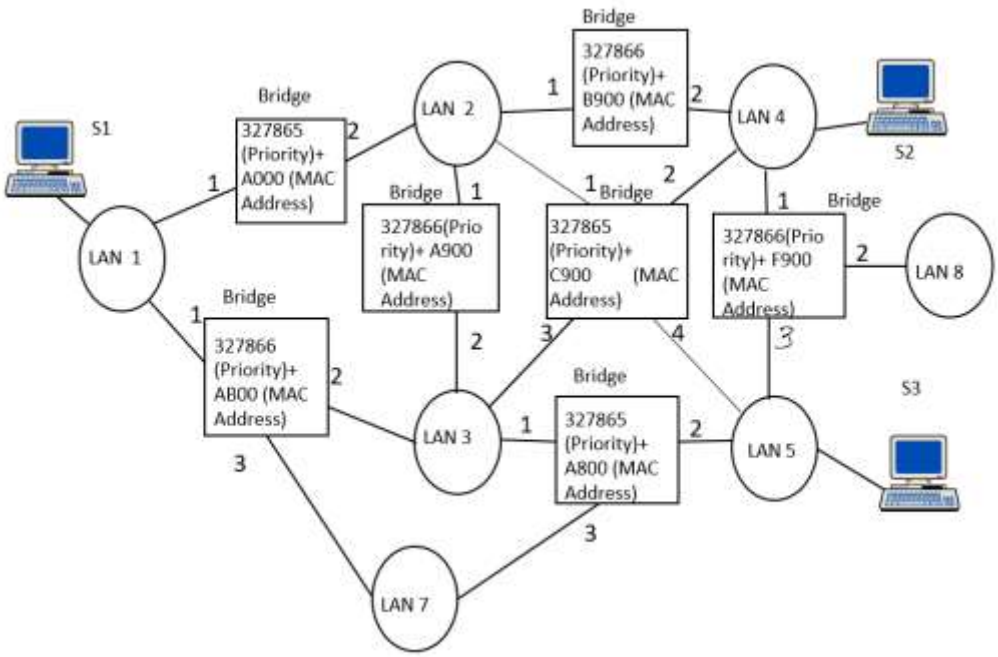
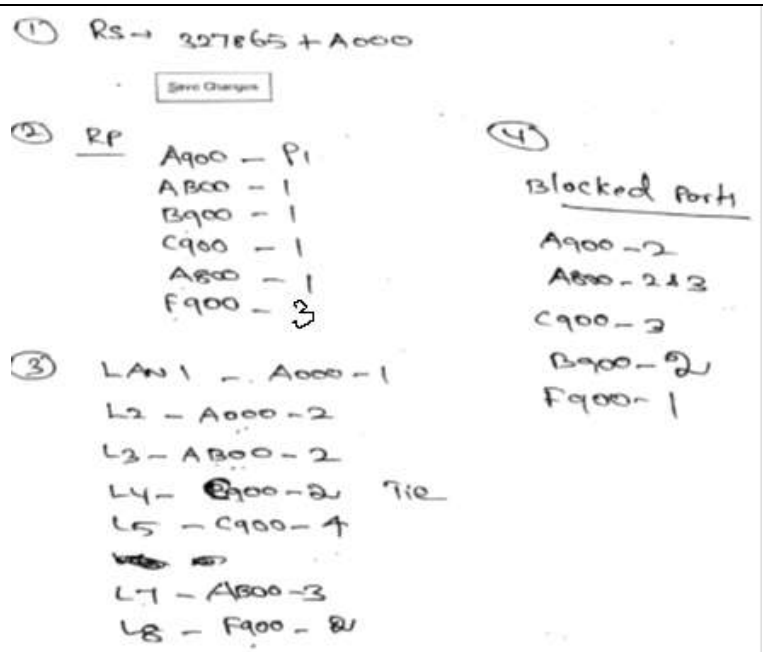
School of Electronics Engineering
Fall Semester 2023-24
Continuous Assessment Test - I

Programme : B.Tech (ECE)
Course Code : BECE401L
Course Name : Computer Communications and Networks
Faculty Name : P PRAKASAM
Slot : C1+TC1/C2+TC2
Class ID : VL2023240100284, VL2023240100285
Date of Exam : 29.05.2023

Max 50 Marks

Q.No	Question	Max Marks
1	i. Match the following services to one or more layers of the OSI model a) Communicates directly with user's application program Application Layer b) Providing independence from different data representation Presentation Layer c) Determining which route through a subnet to choose Network Layer d) Responsibility for carrying frames between adjacent nodes Data Link Layer e) Electrical and Mechanical Interface Physical Layer f) Log-in and log-out procedures Session Layer ii. Suppose a computer sends a packet at the network layer to another computer somewhere in the Internet. The logical destination address of the packet is corrupted. What happens to the packet? How can the source computer be informed of the situation? Before using the destination address in an intermediate or the destination node, the packet goes through error checking that may help the node find the corruption (with a high probability) and discard the packet. Normally the upper layer protocol will inform the source to resend the packet.	6+4
2	Write the Spanning tree algorithm. Compute and draw the spanning tree for the bridged LAN shown in the following figure. Assume that LANs 1, 3 and 5 are Gigabit Ethernets (Cost=15) and the rest are Fast Ethernets (Cost=19). NOTE 1: MAC address is represented in hexadecimal form. Numerical values 1, 2, 3, etc. indicate the port number of each Bridge.  Algorithm (3 marks)	10



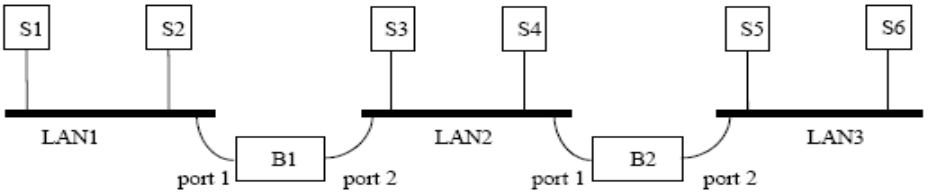
3 Six stations (S1-S6) are connected to an extended LAN through transparent bridges (B1 and B2), as shown in Figure. Initially, the forwarding tables are empty. Suppose the following stations transmit frames: 10

a. S1 transmits to S5
b. S6 transmits to S1
c. S2 transmits to S5
d. S2 transmits to S1
e. S5 transmits to S6

Let station S2 is moved to LAN2

f. S3 transmits to S4
g. S2 transmits to S1
h. S5 transmits to S3
i. S6 transmits to S2
j. S4 transmits to S2

Fill in the forwarding tables with appropriate entries and function after the frames have been completely transmitted**.**



Frame Transmission	B1			B2		
	Address	Port No	Action	Address	Port No	Action
S1-> S5	S1	1	Flood	S1	1	Flood
S6-> S1	S6	2	Forward	S6	2	Forward
S2-> S5	S2	1	Flood	S2	1	Flood
S2-> S1	-	-	Discard	-	-	-
S5-> S6	-	-	-	S5	2	Discard
S2 is moved to LAN2						
S3-> S4	S3	2	Flood	S3	1	Flood
S2-> S1	S2	2	Forward	-	-	Discard
S5-> S3	S5	2	Discard	-	-	Forward
S6-> S2	-	-	Discard	-	-	Forward
S4-> S2	S4	2	Discard	S4	1	Discard

4	i. Suppose that sita@source.org wants to send an email to ram@destination.org that is far away from the sender computer. Draw and explain the typical email operation. Diagram – 4 marks Explanation – 3 marks ii. Find the given domain is either generic or country specific or inverse domain. a). xxx.gov.au – Country Specific b). yyy.gov.in – Country Specific c). 132.128.10.11 – Inverse d). zzz.int – Generic e). 192.145.129.255 – Inverse f). yyy.edu – Generic	10
5	Determine the total delay (latency) for a frame of size 5 million bits that is being sent on a link with 10 routers each having a queuing time of 2 µs and a processing time of 1 µs. The length of the link is 2000 Km. The speed of light inside the link is 2 x *108 mls.* The link has a bandwidth of 5 Mbps. Latency = processing time + queuing time + transmission time + propagation time. Processing time = 10 × 1 µs = 10 µs = 0.000010 s. Queuing time = 10 × 2 µs = 20 µs = 0.000020 s. Transmission time = 5,000,000 / (5 Mbps) = 1 s. Propagation time = (2000 Km) / (2 108) = 0.01 s. Latency = 0.000010 + 0.000020 + 1 + 0.01 = 1.010030 s.	5